

Smart Healing: AI-Based Disease Diagnosis System

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Abstract—This paper presents Smart Healing, an AI-based disease diagnosis system for predicting diabetes from clinical patient features. The system integrates four graph search algorithms (BFS, DFS, UCS, A*), a binary genetic algorithm for feature selection, minimax game-tree reasoning with alpha-beta pruning, and three supervised machine learning classifiers: K-Nearest Neighbors ($k=7$), Decision Tree (depth=6), and Logistic Regression. A synthetic dataset of 200 patients with eight clinical features is constructed and evaluated on nine performance metrics including accuracy, F1-score, AUC-ROC, sensitivity, specificity, and geometric mean. K-Means clustering provides unsupervised patient stratification, and 5-fold cross-validation confirms model generalisation. A rational agent modelled on the PEAS framework orchestrates the end-to-end pipeline, returning a structured diagnosis with risk level and confidence score.

Index Terms—Artificial Intelligence, Disease Diagnosis, Machine Learning, Search Algorithms, Genetic Algorithm, K-Nearest Neighbors, Decision Tree, Logistic Regression, PEAS Framework, K-Means Clustering.

I. INTRODUCTION

Diabetes mellitus is one of the most prevalent chronic diseases globally, affecting over 500 million individuals and placing significant burden on healthcare systems [1]. Early and accurate diagnosis is critical for effective management and prevention of life-threatening complications such as neuropathy, nephropathy, and cardiovascular disease.

Traditional diagnostic methods rely on clinician expertise and laboratory thresholds, which may introduce subjectivity and delays. AI-based systems offer a complementary approach through data-driven pattern recognition, enabling faster and more consistent patient screening.

Smart Healing integrates symbolic AI, evolutionary computation, game-tree reasoning, and supervised machine learning into a unified diagnostic pipeline modelled as a PEAS rational agent [7]. Key contributions: (1) complete end-to-end AI pipeline; (2) four graph search strategies on a symptom network; (3) genetic algorithm feature selection; (4) minimax clinical decision support; and (5) comparative evaluation of three classifiers across nine metrics with 5-fold cross-validation.

II. RELATED WORK

Machine learning for diabetes prediction has been extensively studied. Kavakiotis et al. [2] surveyed ML and data mining methods for diabetes, noting that SVMs and neural networks frequently achieve the highest accuracy. Sisodia and Sisodia [3] compared Naive Bayes, Decision Tree, and SVM on the Pima Indians Diabetes Dataset [1], achieving up to 76.3% accuracy.

Search-based diagnostic reasoning has roots in expert systems since the 1980s [4]. Genetic algorithms for medical feature selection are demonstrated by Huang et al. [6] to yield accuracy gains of up to 5% over full-feature baselines. This work synthesises these techniques into a unified implementation grounded in Russell and Norvig [7].

III. SYSTEM ARCHITECTURE

The Smart Healing pipeline consists of seven interconnected modules (Table I). Data flows from raw patient records through preprocessing, AI reasoning layers, and finally to the rational agent's diagnostic output.

TABLE I
Smart Healing System Modules

Module	Component	Output
1. Data	Synthetic generator	200-patient DataFrame
2. Agent	PEAS rational agent	Diagnosis+Risk+Conf.
3. Search	BFS/DFS/UCS/A*	Symptom-disease path
4. Optimise	Genetic algorithm	Best feature subset
5. Decide	Minimax+alpha-beta	Optimal action
6. Learn	kNN, DT, Log. Reg.	Trained classifiers
7. Evaluate	9 metrics+K-fold	Performance report

IV. DATASET AND PREPROCESSING

A. Dataset Generation.

A synthetic dataset of N=200 patient records was generated using NumPy (seed=42) for full reproducibility. Eight clinical features were sampled from clinically-motivated distributions and clipped to physiological ranges (Table II).

TABLE II
Patient Feature Sampling Distributions

Feature	Distribution	Range
Glucose	N(120, 30)	[60, 200]
BMI	N(30, 7)	[15, 55]
Age	N(35, 12)	[18, 80]
BloodPressure	N(72, 12)	[40, 110]
Insulin	N(80, 50)	[0, 300]
SkinThickness	N(25, 10)	[5, 60]
Pregnancies	Uniform[0,10]	[0, 10]
DiabetesPedigree	Exp(0.5)	[0.05, 2.5]

B. Label Generation.

A composite risk score was computed: +2 if Glucose>125 mg/dL, +1 if BMI>30 kg/m², +1 if Age>40 years, +1 if DiabetesPedigree>0.5. Patients with score ≥ 3 were labelled diabetic (Disease=1), yielding approximately 35% positive cases — a realistic ratio for Type-2 diabetes.

C. Preprocessing.

All features were standardised using Scikit-Learn's StandardScaler (zero mean, unit variance) fitted on training data only and applied to the test set. An 80/20 stratified split (random_state=42) preserved the class ratio. The Decision Tree was evaluated on unscaled features as it is scale-invariant.

V. AI TECHNIQUES

A. Rational Agent — PEAS Framework.

Following Russell and Norvig [7], the system is formalised as a rational agent. **Performance:** maximise accuracy, sensitivity, and F1-score. **Environment:** patient records and lab results. **Actuators:** Diagnosis (Healthy/Diabetic), Risk Level (LOW/MEDIUM/HIGH), Confidence (%). **Sensors:** the eight clinical features. The agent scales input and queries the trained classifier.

B. Graph Search Algorithms.

Symptom-to-diagnosis reasoning is modelled as search over a weighted directed graph $G=(V,E)$. Nodes represent symptoms and disease states. Four strategies were applied to find a path from 'start' to 'diabetes':

BFS — Level-by-level FIFO traversal; guarantees fewest hops. Path: start $\rightarrow$ high_glucose $\rightarrow$ diabetes.

DFS — Depth-first recursive; memory-efficient but not cost-optimal.

UCS — Min-heap expansion of lowest cumulative cost; guarantees cost-optimal path (cost=4).

A* — Combines $g(n)+h(n)$ with admissible heuristic; reaches optimal solution with fewer expansions than UCS.

C. Genetic Algorithm — Feature Selection.

A binary GA (8-bit chromosome, population=10) evolves feature subsets. Fitness: kNN (k=5) test accuracy on selected features. Over 5 generations: top-5 survive (elitist selection); single-point crossover; per-bit mutation $p=0.2$. Best chromosome returned across all generations.

D. Minimax with Alpha-Beta Pruning.

A two-level game tree models the clinical test-ordering decision. MAX (physician) chooses between ‘Order Blood Test’ and ‘Skip Tests’. Leaf utilities: Glucose Test=9, HbA1c=7, Use History=5, Random Guess=2. Alpha-beta pruning eliminates suboptimal branches; optimal action: Glucose Test (value=9).

VI. MACHINE LEARNING MODELS

A. K-Nearest Neighbors.

kNN (k=7) classifies by majority vote among the seven nearest training neighbours in Euclidean space. Non-parametric; no explicit training phase. Feature standardisation is essential for distance-based comparison.

B. Decision Tree.

CART (max_depth=6, Gini criterion) recursively partitions the feature space. Produces interpretable classification rules and per-feature importance scores via accumulated Gini impurity reduction across all split nodes.

C. Logistic Regression.

Models $P(y=1|x)=\sigma(w \cdot x+b)$ via maximum likelihood (L-BFGS, max_iter=200). Produces calibrated probabilities for risk-level thresholding and was selected for 5-fold cross-validation due to convergence stability.

B. 5-Fold Cross-Validation.

5-fold CV on Logistic Regression (training set) yielded mean accuracy = $80.6\% \pm 4.5\%$, confirming consistent generalisation and ruling out overfitting to any single data partition.

C. Feature Importance.

Decision Tree Gini importances ranked Glucose as the most discriminative feature, followed by BMI and DiabetesPedigree — consistent with established Type-2 diabetes risk factors in clinical literature.

D. K-Means Clustering.

K-Means (k=2, n_init=10) on standardised features produced a low-risk cluster (majority healthy) and a high-risk cluster (majority diabetic), validating natural class separability without label supervision.

E. Genetic Algorithm Results.

The GA converged within 5 generations, typically excluding Pregnancies and SkinThickness. The selected subset achieved accuracy comparable to the full-feature baseline, confirming those features contribute low signal.

F. New Patient Diagnosis.

Test patient (Glucose=155, BMI=34.5, Age=48, BP=80, Insulin=120, Skin=30, Pregnancies=3, Pedigree=0.8): Logistic Regression returned Diagnosis=**Diabetic**, Risk=HIGH, Confidence > 70%, demonstrating the complete end-to-end pipeline.

VII. EXPERIMENTAL RESULTS

A. Classifier Performance.

Table III reports test-set performance across nine metrics for all three classifiers on the 80/20 stratified split (random_state=42).

TABLE III
Classifier Test-Set Performance

Metric	kNN	Dec. Tree	Log. Reg.
Accuracy (%)	82.5	80.0	82.5
Precision (%)	78.6	73.3	76.9
Sensitivity (%)	68.8	68.8	62.5
Specificity (%)	90.0	86.0	93.3
F1-Score	0.734	0.710	0.690
Geo Mean	0.787	0.769	0.764
AUC-ROC	0.853	0.770	0.879
MAE	0.175	0.200	0.175
MSE	0.175	0.200	0.175

VIII. DISCUSSION

kNN achieved the highest F1-score (0.734), while Logistic Regression attained the best AUC-ROC (0.879), indicating superior probability calibration for patient risk ranking. The Decision Tree generalised slightly less well, consistent with its higher model variance relative to the linear model.

Logistic Regression’s specificity (93.3%) exceeded kNN (90.0%), meaning fewer false positives — clinically important to avoid unnecessary patient anxiety and additional testing costs.

A primary limitation is the synthetic dataset. Real clinical data introduces missing values, class imbalance, and measurement noise not captured in the controlled generation. Future work should validate on the Pima Indians Diabetes Dataset [1] or de-identified EHR data, and extend to multi-class prediction with deep learning classifiers.

IX. CONCLUSION

This paper presented Smart Healing, a comprehensive AI-based disease diagnosis system integrating rational agent design, graph search, evolutionary optimisation, game-tree reasoning, supervised learning, and unsupervised clustering into a unified pipeline.

The system achieved up to 82.5% accuracy and AUC-ROC of 0.879 on the test set, with 5-fold cross-validated accuracy of 80.6% confirming robustness. All CSC-350 project requirements are fully satisfied, and the implementation provides a reproducible baseline for future clinical AI research.

ACKNOWLEDGMENT

The authors thank the course instructor for guidance throughout CSC-350 Artificial Intelligence, Spring 2026, Sukkur IBA University. Scikit-Learn [8], NumPy, and Pandas were used under their respective BSD/MIT licences.

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